

Intro to Rational exponents (Fractions):

A diagram showing the components of a square root expression. The expression is $\sqrt{x}$. A red arrow points to the '2' (index) with the label 'bottom (also called the index)'. Another red arrow points to the 'x' with the label 'top'. A third red arrow points to the 'x' with the label 'radicand.'. Below the expression, the equivalent rational exponent form is shown in green: $= x^{\frac{1}{2}}$.

Powers with a rational exponent of the form $\frac{1}{n}$

A power involving a rational exponent with numerator 1 and denominator n can be interpreted as the n th root of the base:

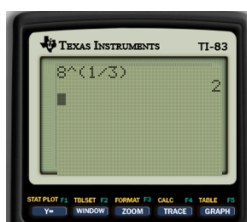
$$b^{\frac{1}{n}} = \sqrt[n]{b}$$

Powers with a rational exponent of the form $\frac{1}{n}$

Example 1: Evaluate each of the following

a) $8^{\frac{1}{3}}$

$= 2$



b) $\sqrt[5]{-32}$

$= (-32)^{\frac{1}{5}}$

$= -2$

c) $-16^{\frac{1}{4}}$

$= -2$

d) $\sqrt[4]{\frac{16}{81}}$

$= \left(\frac{16}{81}\right)^{\frac{1}{4}}$

$= \frac{(16)^{\frac{1}{4}}}{(81)^{\frac{1}{4}}}$

$= \frac{2}{3}$

e) $(-27)^{-\frac{1}{3}}$

$= \frac{1}{(-27)^{\frac{1}{3}}}$

$= \frac{1}{-3}$

$= -\frac{1}{3}$

Powers with a rational exponent of the form $\frac{m}{n}$

You can evaluate a power involving a rational exponent with numerator m and denominator n by taking the n th root of the base raised to the exponent m :

$$b^{\frac{m}{n}} = (\sqrt[n]{b})^m = \sqrt[n]{b^m}$$

Powers with a rational exponent of the form $\frac{m}{n}$

Example 2: Simplify each of the following powers

a)

$$\sqrt[5]{y^2}$$

$$= y^{\frac{2}{5}}$$

b)

$$\sqrt[3]{x}$$

$$= x^{\frac{1}{3}}$$

$$\begin{aligned} \text{c)} \quad & \sqrt{a^{-3}b^{\frac{4}{3}}} \\ &= (a^{-3}b^{\frac{4}{3}})^{1/2} \\ &= a^{-3/2}b^{4/6} \\ &= \frac{b^{2/3}}{a^{3/2}} \end{aligned}$$

$$\begin{aligned} \text{d)} \quad & \sqrt[4]{x^3y^2} \\ &= (x^3y^2)^{1/4} \\ &= x^{3/4}y^{2/4} \\ &= x^{3/4}y^{1/2} \end{aligned}$$

$$\begin{aligned} \text{e)} \quad & \frac{\sqrt[3]{x^2yy^2}}{x^3} \\ &= \frac{(x^2y)^{1/3}y^2}{x^3} \\ &= \frac{x^{2/3}y^{1/3}y^{6/3}}{x^{9/3}} \\ &= x^{-7/3}y^{7/3} \\ &= \frac{y^{7/3}}{x^{7/3}} \end{aligned}$$

Example 3: Evaluate each of the following

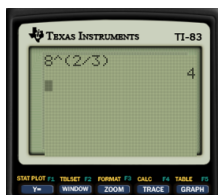
a)

$$8^{\frac{2}{3}}$$

$$= (8^{\frac{1}{3}})^2$$

$$= (2)^2$$

$$= 4$$



b)

$$81^{\frac{5}{4}}$$

$$= (81^{\frac{1}{4}})^5$$

$$= (3)^5$$

$$= 243$$

c)

$$\left(\frac{49}{81}\right)^{-\frac{3}{2}}$$

$$= \left(\frac{81}{49}\right)^{3/2}$$

$$= \frac{(81)^{3/2}}{(49)^{3/2}}$$

$$= \frac{729}{343}$$

If you have a power with a negative exponent and a rational base, just flip the base and make the exponent positive.

Apply Exponent Rules

Example 4: Simplify and express answer using only positive exponents

a)

$$\frac{\left(x^{\frac{2}{3}}\right)\left(x^{\frac{2}{3}}\right)}{\left(x^{\frac{1}{3}}\right)} = \frac{x^{\frac{4}{3}}}{x^{\frac{1}{3}}} = x^{\frac{3}{3}} = x$$

b)

$$\begin{aligned} &\left(y^{\frac{1}{4}}\right)^2 \times \left(y^{-\frac{1}{3}}\right)^2 \\ &= y^{\frac{1}{2}} \times y^{-\frac{2}{3}} \\ &= y^{\frac{3}{6}} \times y^{-\frac{4}{6}} \\ &= y^{-\frac{1}{6}} \\ &= \frac{1}{y^{\frac{1}{6}}} \end{aligned}$$

c)

$$\left(5x^{\frac{1}{2}}\right)^2 \times 4x^{-\frac{1}{2}}$$

$$= 25x \cdot 4x^{-\frac{1}{2}}$$

$$= 25x^{\frac{2}{2}} \cdot 4x^{-\frac{1}{2}}$$

$$= 100x^{\frac{1}{2}}$$

d)

$$\frac{(m^{-2})^3 \sqrt{m^4}}{m \sqrt{pq^{-3}}}$$

$$= \frac{m^{-6} \cdot m^{\frac{4}{2}}}{m p^{\frac{1}{2}} q^{-\frac{3}{2}}}$$

$$= \frac{m^{-6} \cdot m^2}{m p^{\frac{1}{2}} q^{-\frac{3}{2}}}$$

$$= \frac{m^{-4}}{m p^{\frac{1}{2}} q^{-\frac{3}{2}}}$$

$$= \frac{m^{-5}}{p^{\frac{1}{2}} q^{-\frac{3}{2}}}$$

$$= \frac{q^{\frac{3}{2}}}{m^5 p^{\frac{1}{2}}}$$

e)

$$\frac{(x^2)^{-4} \sqrt[5]{y^3}}{y \sqrt{x^{-2} y}}$$

$$= \frac{x^{-8} y^{3/5}}{y x^{-1} y^{1/2}}$$

$$= \frac{x^{-7} y^{3/5}}{y^{3/2}}$$

$$= \frac{x^{-7} y^{6/10}}{y^{15/10}}$$

$$= x^{-7} y^{-9/10}$$

$$= \frac{1}{x^7 y^{9/10}}$$

Complete Worksheet